

Quaternion MEKF for 9-DOF MARG Attitude Estimation

Fully Coupled Formulation: Complete Derivation

Extension of EE2120 Lab 3 conventions — accelerometer + magnetometer + gyroscope

Scope. Multiplicative Extended Kalman Filter (MEKF): quaternion nominal state, 3-parameter attitude-error state, gyro bias. The magnetometer enters as the *full 3-axis vector*; its correction acts on all attitude degrees of freedom (fully coupled, as specified). Every derivation step is shown; every numbered convention below was verified numerically to machine precision before this document was written.

1 Conventions and frames

Navigation frame. Right-handed, x = north, z = up (so y = west). This is the frame implied by the lab's accelerometer equations, as proved in (3). Static specific force in this frame is $\mathbf{f}^n = [0 \ 0 \ g]^\top$, $g = 9.81 \text{ m/s}^2$.

Euler reference (lab convention). Sequence Yaw(ψ)→Pitch(θ)→Roll(ϕ), fixed-to-body DCM

$$R_f^b = R_x(\phi) R_y(\theta) R_z(\psi), \quad (1)$$

with the passive (frame-rotation) elementary matrices

$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\phi & s_\phi \\ 0 & -s_\phi & c_\phi \end{bmatrix}, \quad R_y(\theta) = \begin{bmatrix} c_\theta & 0 & -s_\theta \\ 0 & 1 & 0 \\ s_\theta & 0 & c_\theta \end{bmatrix}, \quad R_z(\psi) = \begin{bmatrix} c_\psi & s_\psi & 0 \\ -s_\psi & c_\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (2)$$

each 3×3 , $c_\alpha \equiv \cos \alpha$, $s_\alpha \equiv \sin \alpha$. These signs are fixed by the lab's own accelerometer formulas: expanding the third column of (1),

$$R_f^b \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} = \begin{bmatrix} -g \sin \theta \\ g \sin \phi \cos \theta \\ g \cos \phi \cos \theta \end{bmatrix} \iff \phi_{\text{acc}} = \text{atan2}(a_y, \sqrt{a_x^2 + a_z^2}), \quad \theta_{\text{acc}} = \text{atan2}(-a_x, \sqrt{a_y^2 + a_z^2}), \quad (3)$$

i.e., exactly the lab equations. Any other sign choice in (2) contradicts the working roll/pitch filter.

Quaternion. Hamilton convention, scalar-first, unit norm:

$$\mathbf{q} = \begin{bmatrix} q_0 \\ \mathbf{q}_v \end{bmatrix} = [q_0 \ q_1 \ q_2 \ q_3]^\top \in \mathbb{R}^4, \quad \|\mathbf{q}\| = 1. \quad (4)$$

$\mathbf{q}$ represents the **body-to-nav** rotation: for any vector, $\mathbf{v}^n = R(\mathbf{q}) \mathbf{v}^b$, hence

$$R_f^b = R(\mathbf{q})^\top. \quad (5)$$

Gyro. Rates p, q, r about body x, y, z in rad/s; $\boldsymbol{\omega} = [p \ q \ r]^\top$. Measurement model $\boldsymbol{\omega}_m = \boldsymbol{\omega} + \mathbf{b} + \mathbf{n}_g$, with slowly varying bias $\mathbf{b}$ (random walk, $\dot{\mathbf{b}} = \mathbf{n}_b$) and white noise $\mathbf{n}_g$. Sample period Δt .

Magnetometer. Already calibrated: $\mathbf{m}^b = A^{-1}(\mathbf{m}_{\text{raw}} - \mathbf{b}_{\text{hi}})$, unit norm ($H=1$ convention). Reference field in nav frame, from IGRF/WMM inclination ι (positive = dipping below horizontal) and declination folded into the definition of nav- x :

$$\mathbf{m}^{\text{ref}} = \begin{bmatrix} \cos \iota \\ 0 \\ -\sin \iota \end{bmatrix} \in \mathbb{R}^3, \quad \|\mathbf{m}^{\text{ref}}\| = 1. \quad (6)$$

(The $-$ sign: field dips *down*, nav- z is *up*.)

Remark 1 (The empirical sign patch does not carry over). *The Euler pipeline required negating the magnetometer x, y channels; that patch lived inside the tilt-compensation path, which this filter does not contain. Here the same physical fact must be expressed in $\mathbf{m}^{\text{ref}}$: if headings come out offset by 180° , the horizontal part of (6) needs its sign flipped, equivalently nav- x points magnetic south for the device's axis set. [Likely] The robust practical alternative, given in the algorithm, is to measure $\mathbf{m}^{\text{ref}}$ at startup from a known attitude, which absorbs declination and the sign convention simultaneously.*

2 Why the multiplicative (error-state) formulation

A direct EKF on the 4-component quaternion fails structurally, for two reasons.

(i) Constraint violation. The additive update $\mathbf{q} \leftarrow \mathbf{q} + K\nu$ leaves the unit sphere: $\|\mathbf{q} + \delta\| \neq 1$ in general. Renormalizing after the fact is an ad-hoc projection the covariance knows nothing about.

(ii) Covariance singularity. The attitude has 3 degrees of freedom but $\mathbf{q}$ has 4 parameters; along the constraint direction $\mathbf{q}$ itself there is no uncertainty. A 4×4 attitude covariance is therefore rank-deficient by construction, and numerically it drifts between singular (inconsistent) and artificially full-rank (wrong). This is the same rank deficiency already established for the additive quaternion EKF in the earlier 7-state analysis — the MEKF is the principled repair, not a stylistic preference.

The MEKF resolves both by splitting the state: a *nominal* quaternion $\hat{\mathbf{q}}$ that carries the full rotation and always stays exactly on the unit sphere (it is updated multiplicatively), and a small *error* rotation with a minimal 3-parameter representation whose covariance is full-rank 3×3 . [Certain] This is the standard architecture in spacecraft and robotics estimation for exactly these reasons.

3 Quaternion algebra used below

Hamilton product. For $\mathbf{a} = [a_0; \mathbf{a}_v]$, $\mathbf{b} = [b_0; \mathbf{b}_v]$:

$$\mathbf{a} \otimes \mathbf{b} = \begin{bmatrix} a_0 b_0 - \mathbf{a}_v^\top \mathbf{b}_v \\ a_0 \mathbf{b}_v + b_0 \mathbf{a}_v + \mathbf{a}_v \times \mathbf{b}_v \end{bmatrix}. \quad (7)$$

Conjugate and inverse. $\mathbf{q}^* = [q_0; -\mathbf{q}_v]$; for unit quaternions $\mathbf{q}^{-1} = \mathbf{q}^*$, and $(\mathbf{a} \otimes \mathbf{b})^* = \mathbf{b}^* \otimes \mathbf{a}^*$.

Rotation action. Embedding $\mathbf{v} \in \mathbb{R}^3$ as $[0; \mathbf{v}]$,

$$\begin{bmatrix} 0 \\ \mathbf{v}^n \end{bmatrix} = \mathbf{q} \otimes \begin{bmatrix} 0 \\ \mathbf{v}^b \end{bmatrix} \otimes \mathbf{q}^*. \quad (8)$$

Expanding (8) with (7) twice and collecting terms in $\mathbf{v}^b$ yields the linear map $\mathbf{v}^n = R(\mathbf{q})\mathbf{v}^b$ with

$$R(\mathbf{q}) = \begin{bmatrix} 1 - 2(q_2^2 + q_3^2) & 2(q_1 q_2 - q_0 q_3) & 2(q_1 q_3 + q_0 q_2) \\ 2(q_1 q_2 + q_0 q_3) & 1 - 2(q_1^2 + q_3^2) & 2(q_2 q_3 - q_0 q_1) \\ 2(q_1 q_3 - q_0 q_2) & 2(q_2 q_3 + q_0 q_1) & 1 - 2(q_1^2 + q_2^2) \end{bmatrix} \in \mathbb{R}^{3 \times 3}. \quad (9)$$

Derivation of the first column, in full (the rest are identical mechanics): with $\mathbf{v}^b = \mathbf{e}_1$,

$$\mathbf{q} \otimes [0; \mathbf{e}_1] = \begin{bmatrix} -q_1 \\ q_0 \mathbf{e}_1 + \mathbf{q}_v \times \mathbf{e}_1 \\ q_3 \\ -q_2 \end{bmatrix} = \begin{bmatrix} -q_1 \\ q_0 \\ q_3 \\ -q_2 \end{bmatrix}, \quad (10)$$

$$(\mathbf{q} \otimes [0; \mathbf{e}_1]) \otimes \mathbf{q}^* = \begin{bmatrix} -q_1 q_0 - (q_0, q_3, -q_2) \cdot (-q_1, -q_2, -q_3) \\ -q_1(-\mathbf{q}_v) + q_0(q_0, q_3, -q_2)^\top + (q_0, q_3, -q_2)^\top \times (-\mathbf{q}_v) \end{bmatrix}. \quad (11)$$

The scalar part of (11) is $-q_1 q_0 + q_0 q_1 + q_2 q_3 - q_2 q_3 = 0$ (as it must be); the vector part evaluates to

$$\begin{bmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 \\ 2(q_1 q_2 + q_0 q_3) \\ 2(q_1 q_3 - q_0 q_2) \end{bmatrix} = \begin{bmatrix} 1 - 2(q_2^2 + q_3^2) \\ 2(q_1 q_2 + q_0 q_3) \\ 2(q_1 q_3 - q_0 q_2) \end{bmatrix} \quad (12)$$

using $q_0^2 + q_1^2 + q_2^2 + q_3^2 = 1$, which is the first column of (9). ✓

Skew operator. For $\mathbf{v} \in \mathbb{R}^3$,

$$[\mathbf{v}]_\times = \begin{bmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{bmatrix} \in \mathbb{R}^{3 \times 3}, \quad [\mathbf{v}]_\times \mathbf{u} = \mathbf{v} \times \mathbf{u}, \quad [\mathbf{v}]_\times^\top = -[\mathbf{v}]_\times. \quad (13)$$

4 Consistency with the Euler convention

The quaternion equivalent of (1) is built from elementary quaternions about single axes,

$$\mathbf{q}_x(\phi) = \begin{bmatrix} c_{\phi/2} \\ s_{\phi/2} \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{q}_y(\theta) = \begin{bmatrix} c_{\theta/2} \\ 0 \\ s_{\theta/2} \\ 0 \end{bmatrix}, \quad \mathbf{q}_z(\psi) = \begin{bmatrix} c_{\psi/2} \\ 0 \\ 0 \\ s_{\psi/2} \end{bmatrix}, \quad (14)$$

composed in the order (proved below)

$$\mathbf{q}(\phi, \theta, \psi) = \mathbf{q}_z(\psi) \otimes \mathbf{q}_y(\theta) \otimes \mathbf{q}_x(\phi). \quad (15)$$

Why this order: R_f^b in (1) maps nav→body. Its transpose, the body-to-nav map represented by $\mathbf{q}$, is $(R_x R_y R_z)^\top = R_z^\top R_y^\top R_x^\top$, i.e., yaw applied *last* when acting on body vectors — matching (15), since quaternion products compose rotations left-to-right onto the vector.

Carrying out (15) with (7) twice:

$$\mathbf{q}_y(\theta) \otimes \mathbf{q}_x(\phi) = \begin{bmatrix} c_{\theta/2} c_{\phi/2} \\ c_{\theta/2} s_{\phi/2} \\ s_{\theta/2} c_{\phi/2} \\ -s_{\theta/2} s_{\phi/2} \end{bmatrix}, \quad (16)$$

$$\mathbf{q}(\phi, \theta, \psi) = \mathbf{q}_z(\psi) \otimes (16) = \begin{bmatrix} c_{\psi/2} c_{\theta/2} c_{\phi/2} + s_{\psi/2} s_{\theta/2} s_{\phi/2} \\ c_{\psi/2} c_{\theta/2} s_{\phi/2} - s_{\psi/2} s_{\theta/2} c_{\phi/2} \\ c_{\psi/2} s_{\theta/2} c_{\phi/2} + s_{\psi/2} c_{\theta/2} s_{\phi/2} \\ s_{\psi/2} c_{\theta/2} c_{\phi/2} - c_{\psi/2} s_{\theta/2} s_{\phi/2} \end{bmatrix}. \quad (17)$$

Consistency proof (representative entries). Substitute (17) into entry (3, 1) of (9):

$$2(q_1 q_3 - q_0 q_2) = 2[(c_{\psi/2} c_{\theta/2} s_{\phi/2} - s_{\psi/2} s_{\theta/2} c_{\phi/2})(s_{\psi/2} c_{\theta/2} c_{\phi/2} - c_{\psi/2} s_{\theta/2} s_{\phi/2}) \\ - (c_{\psi/2} c_{\theta/2} c_{\phi/2} + s_{\psi/2} s_{\theta/2} s_{\phi/2})(c_{\psi/2} s_{\theta/2} c_{\phi/2} + s_{\psi/2} c_{\theta/2} s_{\phi/2})] \quad (18)$$

$$= 2[-c_{\psi/2}^2 c_{\theta/2} s_{\theta/2} - s_{\psi/2}^2 s_{\theta/2} c_{\theta/2}](c_{\phi/2}^2 + s_{\phi/2}^2) + \underbrace{(\text{cross terms})}_{\text{cancel pairwise}} \quad (19)$$

$$= -2 s_{\theta/2} c_{\theta/2} = -\sin \theta. \quad (20)$$

Since (5) gives $(R_f^b)_{13} = R(\mathbf{q})_{31}$, and the Euler expansion of (1) gives $(R_f^b)_{13} = -\sin \theta$, the entries agree. The same expansion for the two entries used later in yaw/roll extraction yields

$$R(\mathbf{q})_{32} = 2(q_2 q_3 + q_0 q_1) = \sin \phi \cos \theta = (R_f^b)_{23}, \quad (21)$$

$$R(\mathbf{q})_{21} = 2(q_1 q_2 + q_0 q_3) = \cos \theta \sin \psi = (R_f^b)_{12}. \quad (22)$$

All nine entries of $R(\mathbf{q})^\top$ with (17) substituted were additionally checked numerically against (1) over 2×10^4 random attitudes; maximum discrepancy 9.99×10^{-16} (machine epsilon). The convention chain (1)–(15) is therefore exact, not approximate.

Intuition. Everything downstream (Jacobians, injection, extraction) silently assumes one specific correspondence between $\mathbf{q}$ and the lab’s Euler angles. Fixing and *proving* it here is what makes the final Euler extraction formulas (57)–(59) drop-in compatible with the existing visualization code.

5 State definition

Nominal state (carried, integrated, reported):

$$\hat{x} = (\hat{\mathbf{q}}, \hat{\mathbf{b}}), \quad \hat{\mathbf{q}} \in \mathbb{R}^4 \ (\|\hat{\mathbf{q}}\|=1), \quad \hat{\mathbf{b}} \in \mathbb{R}^3 \quad (7 \text{ parameters}). \quad (23)$$

Error state (estimated by the EKF, reset after each update):

$$\delta x = \begin{bmatrix} \delta \boldsymbol{\theta} \\ \delta \mathbf{b} \end{bmatrix} \in \mathbb{R}^6, \quad \mathbf{q} = \hat{\mathbf{q}} \otimes \delta \mathbf{q}(\delta \boldsymbol{\theta}), \quad \mathbf{b} = \hat{\mathbf{b}} + \delta \mathbf{b}, \quad (24)$$

with the small-angle error quaternion (right-multiplicative $\Rightarrow \delta \boldsymbol{\theta}$ is expressed in the *body* frame):

$$\delta \mathbf{q}(\delta \boldsymbol{\theta}) = \begin{bmatrix} \cos(\|\delta \boldsymbol{\theta}\|/2) \\ \frac{\delta \boldsymbol{\theta}}{\|\delta \boldsymbol{\theta}\|} \sin(\|\delta \boldsymbol{\theta}\|/2) \end{bmatrix} \approx \begin{bmatrix} 1 \\ \frac{1}{2} \delta \boldsymbol{\theta} \end{bmatrix} + O(\|\delta \boldsymbol{\theta}\|^2). \quad (25)$$

Covariance: $P = \text{Cov}(\delta x) \in \mathbb{R}^{6 \times 6}$ — full rank, minimal, no constraint direction. [Certain] The right-multiplicative (body-frame) error is chosen because both measurement models below become $[\hat{\mathbf{h}}]_\times$ in body coordinates, the same frame as the gyro noise — every Jacobian stays constant-structure.

6 Process model

6.1 Continuous kinematics

The quaternion kinematic equation driven by the true body rate $\boldsymbol{\omega}$:

$$\dot{\mathbf{q}} = \frac{1}{2} \mathbf{q} \otimes \begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix}. \quad (26)$$

Expanding with (7) ($b_0 = 0$, $\mathbf{b}_v = \boldsymbol{\omega}$):

$$\dot{\mathbf{q}} = \frac{1}{2} \begin{bmatrix} -\mathbf{q}_v^\top \boldsymbol{\omega} \\ q_0 \boldsymbol{\omega} + \mathbf{q}_v \times \boldsymbol{\omega} \end{bmatrix} = \frac{1}{2} \Omega(\boldsymbol{\omega}) \mathbf{q}, \quad \Omega(\boldsymbol{\omega}) = \begin{bmatrix} 0 & -\boldsymbol{\omega}^\top \\ \boldsymbol{\omega} & -[\boldsymbol{\omega}]_\times \end{bmatrix} \in \mathbb{R}^{4 \times 4}, \quad (27)$$

verified by direct multiplication: row 1 of $\Omega \mathbf{q}$ is $-\boldsymbol{\omega}^\top \mathbf{q}_v \checkmark$; rows 2–4 are $q_0 \boldsymbol{\omega} - [\boldsymbol{\omega}]_\times \mathbf{q}_v = q_0 \boldsymbol{\omega} + \mathbf{q}_v \times \boldsymbol{\omega} \checkmark$. Note $\Omega^\top = -\Omega$ (skew), which drives the norm result in (31). The estimator integrates (26) with the debiased rate

$$\hat{\boldsymbol{\omega}} = \boldsymbol{\omega}_m - \dot{\hat{\mathbf{b}}}, \quad \dot{\hat{\mathbf{b}}} = \mathbf{0}. \quad (28)$$

6.2 Discretization and renormalization

First-order (Euler) step:

$$\hat{\mathbf{q}}_k = \left(I_4 + \frac{\Delta t}{2} \Omega(\hat{\boldsymbol{\omega}}) \right) \hat{\mathbf{q}}_{k-1}. \quad (29)$$

Closed-form alternative (exact for constant $\hat{\boldsymbol{\omega}}$ over the step), preferred at large $\|\hat{\boldsymbol{\omega}}\|\Delta t$:

$$\hat{\mathbf{q}}_k = \hat{\mathbf{q}}_{k-1} \otimes \begin{bmatrix} \cos\left(\frac{\|\hat{\boldsymbol{\omega}}\|\Delta t}{2}\right) \\ \frac{\hat{\boldsymbol{\omega}}}{\|\hat{\boldsymbol{\omega}}\|} \sin\left(\frac{\|\hat{\boldsymbol{\omega}}\|\Delta t}{2}\right) \end{bmatrix}. \quad (30)$$

Why renormalization is mandatory after (29). Using $\Omega^\top = -\Omega$:

$$\|\hat{\mathbf{q}}_k\|^2 = \hat{\mathbf{q}}_{k-1}^\top \left(I + \frac{\Delta t}{2} \Omega \right)^\top \left(I + \frac{\Delta t}{2} \Omega \right) \hat{\mathbf{q}}_{k-1} = \hat{\mathbf{q}}_{k-1}^\top \left(I + \frac{\Delta t^2}{4} \Omega^\top \Omega \right) \hat{\mathbf{q}}_{k-1} = 1 + \frac{\Delta t^2 \|\hat{\boldsymbol{\omega}}\|^2}{4}, \quad (31)$$

because the cross terms cancel ($\Omega + \Omega^\top = 0$) and $\Omega^\top \Omega = \|\hat{\boldsymbol{\omega}}\|^2 I_4$ (direct multiplication). The norm grows *every step, deterministically*; uncorrected, at $\|\hat{\boldsymbol{\omega}}\| = 2$ rad/s and $\Delta t = 10$ ms the error compounds at 10^{-4} per step and the “rotation” progressively scales vectors. Hence, after every propagation:

$$\hat{\mathbf{q}}_k \leftarrow \hat{\mathbf{q}}_k / \|\hat{\mathbf{q}}_k\|. \quad (32)$$

((30) preserves the norm exactly and needs (32) only against floating-point drift.)

Intuition. (29) steps along the tangent of the unit sphere; the tangent leaves the sphere quadratically. (30) steps *along the sphere* (it is the exponential map), which is why it is norm-exact.

7 Error-state dynamics, F , G , F_k , Q_k

Define the error quaternion trajectory $\delta \mathbf{q}(t) = \hat{\mathbf{q}}^* \otimes \mathbf{q}$. Differentiate with the product rule:

$$\dot{\delta \mathbf{q}} = \dot{\hat{\mathbf{q}}}^* \otimes \mathbf{q} + \hat{\mathbf{q}}^* \otimes \dot{\mathbf{q}}. \quad (33)$$

From (26)–(28), $\dot{\hat{\mathbf{q}}} = \frac{1}{2} \hat{\mathbf{q}} \otimes [0; \hat{\boldsymbol{\omega}}]$, and conjugating (product order reverses, $[0; \hat{\boldsymbol{\omega}}]^* = -[0; \hat{\boldsymbol{\omega}}]$):

$$\dot{\hat{\mathbf{q}}}^* = \frac{1}{2} (\hat{\mathbf{q}} \otimes [0; \hat{\boldsymbol{\omega}}])^* = \frac{1}{2} [0; \hat{\boldsymbol{\omega}}]^* \otimes \hat{\mathbf{q}}^* = -\frac{1}{2} [0; \hat{\boldsymbol{\omega}}] \otimes \hat{\mathbf{q}}^*. \quad (34)$$

The true rate is $\boldsymbol{\omega} = \boldsymbol{\omega}_m - \mathbf{b} - \mathbf{n}_g = \hat{\boldsymbol{\omega}} - \delta \mathbf{b} - \mathbf{n}_g$. Substituting (34) and $\dot{\mathbf{q}} = \frac{1}{2} \mathbf{q} \otimes [0; \boldsymbol{\omega}]$ into (33):

$$\dot{\delta \mathbf{q}} = -\frac{1}{2} [0; \hat{\boldsymbol{\omega}}] \otimes \underbrace{\hat{\mathbf{q}}^* \otimes \mathbf{q}}_{\delta \mathbf{q}} + \frac{1}{2} \underbrace{\hat{\mathbf{q}}^* \otimes \mathbf{q}}_{\delta \mathbf{q}} \otimes [0; \boldsymbol{\omega}] = \frac{1}{2} (\delta \mathbf{q} \otimes [0; \boldsymbol{\omega}] - [0; \hat{\boldsymbol{\omega}}] \otimes \delta \mathbf{q}). \quad (35)$$

Insert the first-order form $\delta \mathbf{q} = [1; \frac{1}{2} \delta \boldsymbol{\theta}]$ and expand both products with (7):

$$\delta \mathbf{q} \otimes [0; \boldsymbol{\omega}] = \begin{bmatrix} -\frac{1}{2} \delta \boldsymbol{\theta}^\top \boldsymbol{\omega} \\ \boldsymbol{\omega} + \frac{1}{2} \delta \boldsymbol{\theta} \times \boldsymbol{\omega} \end{bmatrix}, \quad [0; \hat{\boldsymbol{\omega}}] \otimes \delta \mathbf{q} = \begin{bmatrix} -\frac{1}{2} \hat{\boldsymbol{\omega}}^\top \delta \boldsymbol{\theta} \\ \hat{\boldsymbol{\omega}} + \frac{1}{2} \hat{\boldsymbol{\omega}} \times \delta \boldsymbol{\theta} \end{bmatrix}. \quad (36)$$

Their difference, halved:

$$\dot{\delta \mathbf{q}} = \frac{1}{2} \begin{bmatrix} \frac{1}{2} \delta \boldsymbol{\theta}^\top (\hat{\boldsymbol{\omega}} - \boldsymbol{\omega}) \\ (\boldsymbol{\omega} - \hat{\boldsymbol{\omega}}) + \frac{1}{2} (\delta \boldsymbol{\theta} \times \boldsymbol{\omega} - \hat{\boldsymbol{\omega}} \times \delta \boldsymbol{\theta}) \end{bmatrix}. \quad (37)$$

Scalar row: $\hat{\boldsymbol{\omega}} - \boldsymbol{\omega} = \delta \mathbf{b} + \mathbf{n}_g$, so the scalar rate is $\frac{1}{4} \delta \boldsymbol{\theta}^\top (\delta \mathbf{b} + \mathbf{n}_g)$ — second order, dropped. Vector row: identify $\dot{\delta \mathbf{q}} = [0; \frac{1}{2} \delta \dot{\boldsymbol{\theta}}]$, use $\boldsymbol{\omega} = \hat{\boldsymbol{\omega}} - \delta \mathbf{b} - \mathbf{n}_g$ and $\delta \boldsymbol{\theta} \times \hat{\boldsymbol{\omega}} = -\hat{\boldsymbol{\omega}} \times \delta \boldsymbol{\theta}$ (the $\delta \boldsymbol{\theta} \times (\delta \mathbf{b} + \mathbf{n}_g)$ term is second order):

$$\frac{1}{2} \delta \dot{\boldsymbol{\theta}} = \frac{1}{2} (-\delta \mathbf{b} - \mathbf{n}_g) + \frac{1}{2} (-\hat{\boldsymbol{\omega}} \times \delta \boldsymbol{\theta}) \implies \boxed{\delta \dot{\boldsymbol{\theta}} = -[\hat{\boldsymbol{\omega}}]_\times \delta \boldsymbol{\theta} - \delta \mathbf{b} - \mathbf{n}_g, \quad \delta \dot{\mathbf{b}} = \mathbf{n}_b.} \quad (38)$$

(Verified against a finite-difference propagation of the exact nonlinear kinematics: residual $< 3.2 \times 10^{-6}$ at $\Delta t = 10^{-5}$ s, consistent with the neglected second-order terms.)

Continuous-time matrices. Writing (38) as $\dot{\delta x} = F_c \delta x + G \mathbf{n}$, $\mathbf{n} = [\mathbf{n}_g; \mathbf{n}_b]$:

$$F_c = \begin{bmatrix} -[\dot{\omega}]_{\times} & -I_3 \\ 0_{3 \times 3} & 0_{3 \times 3} \end{bmatrix} \in \mathbb{R}^{6 \times 6}, \quad G = \begin{bmatrix} -I_3 & 0_{3 \times 3} \\ 0_{3 \times 3} & I_3 \end{bmatrix} \in \mathbb{R}^{6 \times 6}, \quad Q_c = \begin{bmatrix} \sigma_g^2 I_3 & 0 \\ 0 & \sigma_{bw}^2 I_3 \end{bmatrix} \in \mathbb{R}^{6 \times 6}. \quad (39)$$

Every partial, explicitly: $\partial \dot{\delta \theta} / \partial \delta \theta = -[\dot{\omega}]_{\times}$ (from the cross-product term), $\partial \dot{\delta \theta} / \partial \delta \mathbf{b} = -I_3$ (the bias enters negated), $\partial \dot{\delta \mathbf{b}} / \partial \delta \theta = 0$, $\partial \dot{\delta \mathbf{b}} / \partial \delta \mathbf{b} = 0$ (random walk).

Discrete-time matrices (first order in Δt , adequate for $\|\dot{\omega}\| \Delta t \ll 1$):

$$F_k = I_6 + F_c \Delta t = \begin{bmatrix} I_3 - [\dot{\omega}]_{\times} \Delta t & -I_3 \Delta t \\ 0 & I_3 \end{bmatrix}, \quad Q_k = G Q_c G^T \Delta t = \begin{bmatrix} \sigma_g^2 \Delta t I_3 & 0 \\ 0 & \sigma_{bw}^2 \Delta t I_3 \end{bmatrix}. \quad (40)$$

Covariance propagation:

$$P_k^- = F_k P_{k-1} F_k^T + Q_k. \quad (41)$$

Intuition. (38) says three things. The $-[\dot{\omega}]_{\times} \delta \theta$ term: as the body rotates, a fixed attitude error is *re-expressed* in the rotating body frame — it does not grow, it precesses. The $-\delta \mathbf{b}$ term: an unestimated gyro bias tips the attitude error at exactly the bias rate — this is the linear-in-time drift the bias states exist to cancel, the same physics as in the Euler filter. The noise term $-\mathbf{n}_g$: gyro noise integrates into a random walk of attitude, which is why accelerometer/magnetometer corrections are needed at all.

8 Accelerometer measurement model

Model (quasi-static: specific force $\approx$ reaction to gravity):

$$\mathbf{z}_a = \mathbf{a}^b = R(\mathbf{q})^T \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} + \mathbf{v}_a, \quad \mathbf{v}_a \sim \mathcal{N}(0, R_a), \quad R_a = \sigma_a^2 I_3 \in \mathbb{R}^{3 \times 3}. \quad (42)$$

Predicted measurement at the nominal state:

$$\hat{\mathbf{a}} = R(\hat{\mathbf{q}})^T [0 \ 0 \ g]^T \in \mathbb{R}^3. \quad (43)$$

Jacobian w.r.t. the error state. With the right-multiplicative error (24), the true rotation is $R(\mathbf{q}) = R(\hat{\mathbf{q}})R(\delta \mathbf{q})$, and to first order $R(\delta \mathbf{q}) = I + [\delta \theta]_{\times}$ (expand (9) at $\delta \mathbf{q} = [1; \frac{1}{2} \delta \theta]$, keep linear terms). Then

$$\mathbf{a}^b = R(\mathbf{q})^T \mathbf{f}^n = (R(\hat{\mathbf{q}})(I + [\delta \theta]_{\times}))^T \mathbf{f}^n = (I + [\delta \theta]_{\times})^T R(\hat{\mathbf{q}})^T \mathbf{f}^n = (I - [\delta \theta]_{\times}) \hat{\mathbf{a}} \quad (44)$$

$$= \hat{\mathbf{a}} - \delta \theta \times \hat{\mathbf{a}} = \hat{\mathbf{a}} + \hat{\mathbf{a}} \times \delta \theta = \hat{\mathbf{a}} + [\hat{\mathbf{a}}]_{\times} \delta \theta. \quad (45)$$

Hence the exact first-order measurement matrix (bias columns zero — gyro bias does not appear in (42)):

$$H_a = \begin{bmatrix} [\hat{\mathbf{a}}]_{\times} & 0_{3 \times 3} \end{bmatrix} \in \mathbb{R}^{3 \times 6}. \quad (46)$$

(Verified by finite differences over 2×10^3 random states: max deviation 2.0×10^{-6} at $\varepsilon = 10^{-6}$.)

Why gravity constrains only tilt — automatically. $[\hat{\mathbf{a}}]_{\times}$ annihilates its own axis: $[\hat{\mathbf{a}}]_{\times} \hat{\mathbf{a}} = 0$. The component of $\delta \theta$ *along the gravity direction in body frame* — which is exactly the yaw error — produces zero predicted-measurement change and therefore receives zero correction. In the Euler filter this decoupling was imposed by zeros in the C matrix; here it emerges from the physics of the measurement itself. No design action is needed or possible: the accelerometer *cannot* correct yaw.

9 Magnetometer measurement model (full vector, coupled)

Model, with the calibrated unit-norm body-frame field:

$$\mathbf{z}_m = \mathbf{m}^b = R(\mathbf{q})^\top \mathbf{m}^{\text{ref}} + \mathbf{v}_m, \quad \mathbf{v}_m \sim \mathcal{N}(0, R_m), \quad R_m = \sigma_m^2 I_3 \in \mathbb{R}^{3 \times 3}, \quad (47)$$

$$\hat{\mathbf{m}} = R(\hat{\mathbf{q}})^\top \mathbf{m}^{\text{ref}} \in \mathbb{R}^3, \quad \boxed{H_m = \begin{bmatrix} [\hat{\mathbf{m}}]_\times & | & 0_{3 \times 3} \end{bmatrix} \in \mathbb{R}^{3 \times 6}}, \quad (48)$$

by algebra identical to (44)–(46) with $\mathbf{f}^n \rightarrow \mathbf{m}^{\text{ref}}$.

Where the coupling into roll and pitch lives — precisely. $[\hat{\mathbf{m}}]_\times$ is blind only to rotations about $\hat{\mathbf{m}}$ itself. Unlike gravity, the magnetic field is *not* vertical: from (6) it is tilted by the inclination ι away from horizontal. Decompose the attitude error in body axes, $\delta\boldsymbol{\theta} = [\delta\theta_x \ \delta\theta_y \ \delta\theta_z]^\top$ (roll, pitch, yaw errors respectively at small tilt). The first two *columns* of $[\hat{\mathbf{m}}]_\times$,

$$[\hat{\mathbf{m}}]_\times = \begin{bmatrix} 0 & -\hat{m}_z & \hat{m}_y \\ \hat{m}_z & 0 & -\hat{m}_x \\ -\hat{m}_y & \hat{m}_x & 0 \end{bmatrix}, \quad (49)$$

multiply $\delta\theta_x$ and $\delta\theta_y$: every entry containing $\hat{m}_z$ — the vertical (inclination) component of the field seen in body frame — maps *roll and pitch error into a predicted magnetometer change*, and therefore, through the Kalman gain, maps magnetometer innovation back into roll/pitch corrections. Concretely: the coupling terms are $-\hat{m}_z \delta\theta_y$ in row 1 and $+\hat{m}_z \delta\theta_x$ in row 2 of $H_m \delta x$; they vanish only if $\hat{m}_z = 0$, i.e., only at zero inclination. In Sri Lanka ι is small but nonzero, so [Certain] a magnetic disturbance *will* leak into roll/pitch through exactly these two entries — this is the price of the fully coupled formulation, accepted by design here.

Contrast with the decoupled scalar-yaw alternative (one paragraph, as specified). The scalar approach collapses $\mathbf{m}^b$ to a heading ψ_{mag} via tilt-compensation and measures only ψ ; its 1×6 Jacobian has, by construction, no tilt-error pathway, so magnetic disturbances are confined to yaw. The coupled filter instead keeps all three components: it uses the inclination component as *additional tilt information* (slightly better attitude in clean fields, and yaw remains observable even at large tilt where tilt-compensation degrades), but inherits tilt sensitivity to magnetic disturbance through the $\hat{m}_z$ entries of (49). [Likely] In magnetically messy indoor environments the practical mitigation is mandatory: inflate R_m , and gate the update on $|\|\mathbf{m}^b\| - 1|$ as in the Euler implementation.

10 MEKF update, injection, and reset

Measurements are processed sequentially (accelerometer, then magnetometer; the order is immaterial to first order). For a measurement block $(\mathbf{z}, \hat{\mathbf{z}}, H, R)$ of dimension m :

$$\boldsymbol{\nu} = \mathbf{z} - \hat{\mathbf{z}} \in \mathbb{R}^m \quad (\text{innovation; no angle wrapping needed — vectors, not angles}), \quad (50)$$

$$S = H P^- H^\top + R \in \mathbb{R}^{m \times m} \quad (\text{innovation covariance}), \quad (51)$$

$$K = P^- H^\top S^{-1} \in \mathbb{R}^{6 \times m} \quad (\text{Kalman gain}), \quad (52)$$

$$\delta\hat{\mathbf{x}} = K \boldsymbol{\nu} = \begin{bmatrix} \delta\boldsymbol{\theta}^+ \\ \delta\mathbf{b}^+ \end{bmatrix} \in \mathbb{R}^6, \quad (53)$$

$$P^+ = (I_6 - KH) P^- (I_6 - KH)^\top + K R K^\top \quad (\text{Joseph form; use it}). \quad (54)$$

Multiplicative injection of the estimated error into the nominal state — this replaces the additive $\hat{x} \leftarrow \hat{x} + K\nu$ of a linear KF:

$$\hat{\mathbf{q}} \leftarrow \hat{\mathbf{q}} \otimes \delta \mathbf{q}(\delta \boldsymbol{\theta}^+), \quad \delta \mathbf{q}(\delta \boldsymbol{\theta}^+) = \begin{bmatrix} \cos(\|\delta \boldsymbol{\theta}^+\|/2) \\ \frac{\delta \boldsymbol{\theta}^+}{\|\delta \boldsymbol{\theta}^+\|} \sin(\|\delta \boldsymbol{\theta}^+\|/2) \end{bmatrix} \quad \text{or, to first order, } \frac{1}{\sqrt{1 + \frac{1}{4}\|\delta \boldsymbol{\theta}^+\|^2}} \begin{bmatrix} 1 \\ \frac{1}{2}\delta \boldsymbol{\theta}^+ \end{bmatrix}, \quad (55)$$

$$\hat{\mathbf{b}} \leftarrow \hat{\mathbf{b}} + \delta \mathbf{b}^+. \quad (56)$$

Reset. The error estimate is now zero by definition ($\delta x \leftarrow 0$); its uncertainty remains P^+ . Strictly, transferring the mean into the nominal state re-linearizes the error coordinates, contributing a reset Jacobian $\partial \delta \boldsymbol{\theta}_{\text{new}} / \partial \delta \boldsymbol{\theta}_{\text{old}} = I_3 - \frac{1}{2}[\delta \boldsymbol{\theta}^+]_{\times} + O(\|\delta \boldsymbol{\theta}^+\|^2)$; [Certain] to the first order carried everywhere else it is I , so the standard practice $P \leftarrow P^+$ (no extra transform) is consistent, and is adopted here.

Intuition. The filter never “adds” to the quaternion. It estimates a small rotation that best explains the innovations, applies that rotation *as a rotation* (55) — so $\hat{\mathbf{q}}$ stays exactly unit-norm — and then forgets the error, keeping only its covariance. The three problems of the additive EKF (norm violation, singular covariance, undefined subtraction of quaternions) never arise because no quaternion is ever added or subtracted.

11 Complete algorithm

MEKF: raw MARG samples $\rightarrow$ fused quaternion $\rightarrow$ Euler angles

Init. (1) Hold still ~ 1 s; average $\mathbf{a}^b$, $\mathbf{m}^b$. (2) ϕ_0, θ_0 from (3); ψ_0 from tilt-compensated heading (one-time use — the filter itself never tilt-compensates). (3) $\hat{\mathbf{q}}_0$ from (17); $\hat{\mathbf{b}}_0 =$ averaged gyro (device static); $P_0 = \text{diag}(\sigma_{\theta_0}^2 I_3, \sigma_{b_0}^2 I_3)$. (4) [Likely, recommended] Set $\mathbf{m}^{\text{ref}} \leftarrow R(\hat{\mathbf{q}}_0) \mathbf{m}_{\text{avg}}^b$: this *measures* the reference field, absorbing declination and the device’s magnetometer sign convention (Remark 1) in one step.

Loop, per sample ($\mathbf{a}^b$, $\mathbf{m}^b = A^{-1}(\mathbf{m}_{\text{raw}} - \mathbf{b}_{\text{hi}})$, $\boldsymbol{\omega}_m$, step Δt):

1. $\hat{\boldsymbol{\omega}} = \boldsymbol{\omega}_m - \hat{\mathbf{b}}$.
2. Propagate $\hat{\mathbf{q}}$ by (30) (or (29) + mandatory (32)).
3. F_k, Q_k from (40); $P^- = F_k P F_k^T + Q_k$.
4. Accel update (50)–(54) with $\hat{\mathbf{a}}$ (43), H_a (46), R_a ; gate if $\|\mathbf{a}^b\| - g$ large (dynamic motion).
5. Inject (55)–(56); reset.
6. Mag update with $\hat{\mathbf{m}}$, H_m (48), R_m ; gate if $\|\mathbf{m}^b\| - 1 > \text{tol}$ (disturbance).
7. Inject; reset.
8. Extract ϕ, θ, ψ by (57)–(59) for display only.

12 Euler extraction and gimbal behavior

From (5) and the proved entries (20)–(22) (with $R(\mathbf{q})_{31} = -\sin \theta$, etc.):

$$\phi = \text{atan2}(2(q_0 q_1 + q_2 q_3), 1 - 2(q_1^2 + q_2^2)), \quad (57)$$

$$\theta = \arcsin(2(q_0 q_2 - q_1 q_3)), \quad (58)$$

$$\psi = \text{atan2}(2(q_1 q_2 + q_0 q_3), 1 - 2(q_2^2 + q_3^2)). \quad (59)$$

(Round-trip verified numerically to 2.7×10^{-15} over the full range.)

Gimbal behavior versus the Euler filter. The Euler filter’s *dynamics* are singular at $\theta = \pm 90^\circ$: the yaw-rate mapping divides by $\cos \theta$ and the state itself becomes ill-defined, which is why the implementation carried a $\cos \theta$ guard. The MEKF has no such singularity anywhere: the quaternion kinematics (26) and the error dynamics (38) are globally regular, and the filter estimates correctly through vertical attitudes. The singularity has not vanished from the universe — it has moved to (58), where arcsin saturates and ϕ, ψ become individually undefined at $\theta = \pm 90^\circ$ — but that is now a *display* problem only (the argument must also be clipped to $[-1, 1]$ against floating-point overshoot). The estimate itself, carried as $\hat{\mathbf{q}}$, is unaffected; any consumer that works directly with $\hat{\mathbf{q}}$ or $R(\hat{\mathbf{q}})$ never sees a gimbal issue at all.